

The Inference-First Picture of Mind, Logic, and Mathematics

A Formal Theory of Consequence, Model-Generation, Grounding, and Artificial Agents

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Abstract

We develop a formal framework grounded in a single primitive: the consequence relation. We define consequence systems, inference systems, and agents axiomatically; we define model-generation, model-reach, and model-testing; and we introduce the grounding coefficient G , which distinguishes self-sourced from derived sensory grounding. We prove a series of theorems characterizing the relationship between logic, model-generation, language, grounding, and cross-agent understanding. The central results are: (1) reference to absent objects requires model-generation; (2) language presupposes model-generation, but not conversely; (3) understanding between agents requires a shared consequence core; (4) agents whose logics are structurally superior cannot be rationally interpreted by lower agents; (5) ungrounded agents have no correspondence test and cannot detect their own errors. We then extend the framework to artificial agents, distinguishing ungrounded, grounded, and embodied systems, and prove that ungrounded symbolic agents occupy Level 5*; that structural hallucination is undetectable by the agent that suffers it; that scaling does not change the grounding coefficient; and that embodiment is a qualitative rather than a quantitative threshold. The framework is naturalistic but not reductive, and all results are conditional on the stated axioms. This paper is purely theoretical; its application to artificial intelligence and to industrial structure is developed in the companion application series *Grounded Agents* (application series).

Contents

I	Foundations	2
1	Introduction	3
2	The Primitive: Consequence	3
3	Determinacy and Exclusion	4
4	Inference Systems and Agents	4
5	Grounding	5
6	Model-Generation	6
7	Model-Testing	6
II	Cognition, Language, and Levels	7

8	The Threshold Theorem	7
9	Language	7
10	Levels of Cognition	8
III	Higher Intelligences and Cross-Level Understanding	9
11	Logical Compatibility	9
12	The Opacity Theorem	10
13	The Three Scenarios Formalized	10
14	The Main Chain	11
IV	Artificial Agents	11
15	The Artificial Case	11
16	The Grounding Gap in Artificial Systems	12
17	Embodiment and Closure	12
18	AI Trajectories and the Opacity Threshold	12
19	Predictions	13
V	Limits and Conclusion	13
20	Open Problems	13
21	Conclusion	14
A	Summary of Definitions	15
B	Summary of Theorems	16
C	Cognition Level Table	16
D	A Note on Method	16

Part I

Foundations

1 Introduction

Two questions motivate this paper.

First: what is the relationship between logic, mathematics, and reality? The standard answers—that mathematics is discovered, invented, or merely formal—each capture something but leave something out. We propose a different starting point: that the fundamental cognitive act is inference, and that logic, mathematics, and science are successive refinements of that act.

Second: what does it take for a mind to understand reality? We argue that the decisive capacity is *model-generation*—the ability to construct representations of what is not immediately present—and that this capacity admits degrees, with profound consequences for what an organism can experience, know, and communicate.

A third thread, developed formally here and applied in the companion papers, is *grounding*: the distinction between contents drawn from an agent’s own sensory history and contents inherited from other agents. We introduce the *grounding coefficient* G to capture this distinction and to make the framework adequate to artificial agents.

This paper is purely theoretical. It proceeds in four parts. Part I lays out the framework. Part II draws out its implications for cognition, language, and levels. Part III treats higher intelligences and the limits of cross-level understanding. Part IV extends the framework to artificial agents—the case that makes the grounding coefficient indispensable. The application of the framework to artificial intelligence, to industrial structure, and to the shape of the coming transition is developed in the companion application series *Grounded Agents* (application series); the present paper is its theoretical foundation.

2 The Primitive: Consequence

We begin with a single primitive: the relation of consequence.

Definition 2.1 (Consequence Relation). Let $\mathcal{C}$ be a non-empty set of *contents*. A *consequence relation* is a binary relation $\vdash \subseteq \mathcal{P}(\mathcal{C}) \times \mathcal{C}$. We write $\Gamma \vdash c$ to mean: the content c follows from the set of contents Γ .

Definition 2.2 (Consequence System). A *consequence system* is a pair $\mathfrak{S} = (\mathcal{C}, \vdash)$ satisfying:

- (C1) **Reflexivity.** If $c \in \Gamma$, then $\Gamma \vdash c$.
- (C2) **Monotonicity.** If $\Gamma \vdash c$ and $\Gamma \subseteq \Delta$, then $\Delta \vdash c$.
- (C3) **Cut.** If $\Gamma \vdash c$ and $\Delta \cup \{c\} \vdash d$, then $\Gamma \cup \Delta \vdash d$.

Remark 2.3. These are the standard structural rules of a Tarski-style consequence relation. We do *not* assume compactness, completeness, or the existence of a deductive apparatus. The framework is deliberately general.

3 Determinacy and Exclusion

The structural rules alone do not capture the conditions of the possibility of thought. We add two axioms.

Axiom 3.1 (Determinacy / Identity). For every $c \in \mathcal{C}$, $\{c\} \vdash c$.

Axiom 3.2 (Exclusion / Non-Contradiction). There is no $c \in \mathcal{C}$ such that $\vdash c$ and $\vdash \neg c$, where $\neg : \mathcal{C} \rightarrow \mathcal{C}$ is a negation operator satisfying $\neg\neg c = c$.

Remark 3.3. Axiom 3.2 applies to the empty premise set. It rules out *global* triviality but permits local inconsistency to be defined derivatively below.

Definition 3.4 (Consistency and Inconsistency). A set $\Gamma \subseteq \mathcal{C}$ is *consistent* if there is no $c \in \mathcal{C}$ with $\Gamma \vdash c$ and $\Gamma \vdash \neg c$. Otherwise Γ is *inconsistent*.

Definition 3.5 (Explosive System). A consequence system $\mathfrak{S}$ is *explosive* if it satisfies:

(C4) **Explosion.** If Γ is inconsistent, then $\Gamma \vdash d$ for every $d \in \mathcal{C}$.

Remark 3.6. Explosion is a *choice*, not a consequence of (C1)–(C3). Paraconsistent logics reject (C4). All subsequent results that invoke explosion are conditional on it.

Proposition 3.7 (Trivialization). *Suppose $\mathfrak{S}$ is explosive. If every axiom of a theory T is false in the sense that T is inconsistent, then every content in the language is derivable from T .*

Proof. Immediate from (C4) applied to the inconsistent set T . □

This formalizes the observation that a system with all-false axioms trivializes.

4 Inference Systems and Agents

Definition 4.1 (Closure Operator). Let $\mathfrak{S} = (\mathcal{C}, \vdash)$ be a consequence system. A map $\sigma : \mathcal{P}(\mathcal{C}) \rightarrow \mathcal{P}(\mathcal{C})$ is a *closure operator* for $\mathfrak{S}$ if:

- (I1) $\Gamma \subseteq \sigma(\Gamma)$,
- (I2) $\sigma(\sigma(\Gamma)) = \sigma(\Gamma)$,
- (I3) $\Gamma \subseteq \Delta \Rightarrow \sigma(\Gamma) \subseteq \sigma(\Delta)$.

We call $\sigma(\Gamma)$ the *inferential closure* of Γ .

Definition 4.2 (Inference System). An *inference system* is a triple $\mathcal{I} = (\mathcal{C}, \vdash, \sigma)$ where $(\mathcal{C}, \vdash)$ is a consequence system and σ is a closure operator for it.

Definition 4.3 (Agent). An *agent* is a tuple $\mathcal{A} = (\mathcal{I}, \mathcal{S}, \mu)$ where:

- $\mathcal{I} = (\mathcal{C}, \vdash, \sigma)$ is an inference system,
- $\mathcal{S}$ is a set of *sensory states*,
- $\mu : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{C})$ is a *grounding map* assigning to each sensory state the contents “given” by that state.

We call $\mu(s)$ the *empirical content* of s .

Definition 4.4 (Belief Set). The *belief set* of agent $\mathcal{A}$ at sensory state s is

$$\mathcal{B}_{\mathcal{A}}(s) = \sigma(\mu(s)).$$

Definition 4.5 (Present-Bound Map). A grounding map μ is *present-bound* if for every $s \in \mathcal{S}$ and every $c \in \sigma(\mu(s))$, the content c contains no *absent object marker*—that is, no symbol whose semantic role is fixed only by reference to objects not occurring in s .

5 Grounding

The framework so far treats the grounding map μ as primitive. But *where* μ comes from matters enormously. A biological organism’s sensory states are its own; a text-trained artificial system’s contents are inherited from human authors. We introduce a third coordinate to capture this.

Definition 5.1 (Self-Sourced Grounding). Agent $\mathcal{A}$ has *self-sourced grounding* if its grounding map $\mu_{\mathcal{A}} : \mathcal{S}_{\mathcal{A}} \rightarrow \mathcal{P}(\mathcal{C})$ is defined on $\mathcal{A}$ ’s own sensory states, and the contents in $\mu_{\mathcal{A}}(s)$ are causally produced by $\mathcal{A}$ ’s interaction with its environment at s .

Definition 5.2 (Derived Grounding). Agent $\mathcal{A}$ has *derived grounding* if there exists another agent $\mathcal{B}$ and a map $\nu : \mathcal{P}(\mathcal{C}) \rightarrow \mathcal{P}(\mathcal{C})$ such that

$$\mu_{\mathcal{A}}(s) = \nu(\mu_{\mathcal{B}}(s'))$$

for all $s \in \mathcal{S}_{\mathcal{A}}$ and some $s' \in \mathcal{S}_{\mathcal{B}}$. We call ν the *derivation map*.

Definition 5.3 (Grounding Coefficient). Let $\mathcal{A}$ be an agent with belief sets as in Definition 4.4. A content c is *grounded in $\mathcal{A}$ ’s own sensory history* if it lies in the inferential closure of the self-sourced empirical contents, i.e. if

$$c \in \sigma\left(\bigcup_{s \in \mathcal{S}} \mu(s)\right)_{\text{self}},$$

where the superscript “self” restricts the union to those states whose contents are causally produced by $\mathcal{A}$ ’s own interaction, in the sense of Definition 5.1. For each $s \in \mathcal{S}$, define

$$G(\mathcal{A}, s) = \frac{|\{c \in \mathcal{B}_{\mathcal{A}}(s) : c \text{ is grounded in } \mathcal{A}\text{'s own sensory history}\}|}{|\mathcal{B}_{\mathcal{A}}(s)|}.$$

The *grounding coefficient* of $\mathcal{A}$ is

$$G(\mathcal{A}) = \inf_{s \in \mathcal{S}} G(\mathcal{A}, s).$$

Remark 5.4. $G(\mathcal{A}) = 1$ for a fully embodied, self-sourced agent. $G(\mathcal{A}) = 0$ for a purely text-trained system with no self-sourced sensory states. Intermediate values arise for hybrid systems. The definition is stated for agents with finite belief sets; for infinite belief sets one takes the corresponding density where it exists.

6 Model-Generation

Definition 6.1 (Model). A *model* for agent $\mathcal{A}$ is a subset $M \subseteq \mathcal{C}$ such that M is consistent and $\sigma(M) = M$.

Definition 6.2 (Model-Generator). Let $T \subseteq \mathcal{C}$ be a set of *target contents*. Agent $\mathcal{A}$ has a *model-generator* with respect to T if there exists a function $\gamma : \mathcal{P}(\mathcal{C}) \rightarrow \mathcal{P}(\mathcal{C})$ such that for every $s \in \mathcal{S}$:

- (M1) $\gamma(\mu(s))$ is a model,
- (M2) if $T \neq \emptyset$, then $\gamma(\mu(s)) \cap T \neq \emptyset$,
- (M3) γ is *extensive*: $\Gamma \subseteq \gamma(\Gamma)$ for all Γ ,
- (M4) γ is *monotone*: $\Gamma \subseteq \Delta \Rightarrow \gamma(\Gamma) \subseteq \gamma(\Delta)$.

Definition 6.3 (Model-Reach). The *model-reach* of γ is

$$R(\gamma) = \sup_{s \in \mathcal{S}} |\gamma(\mu(s)) \setminus \mu(s)|.$$

Remark 6.4. Condition (M3) and Definition 6.3 together capture the “detachment from the present”: a model-generator produces contents beyond what is sensed.

7 Model-Testing

Definition 7.1 (Observation Operator). An *observation operator* for agent $\mathcal{A}$ is a function

$$\omega : \mathcal{P}(\mathcal{C}) \times \mathcal{S} \rightarrow \{\top, \perp\}$$

satisfying:

- (O1) $\omega(M, s) = \top \Rightarrow M \cup \mu(s)$ is consistent,
- (O2) $\omega(M, s) = \perp \Rightarrow M \cup \mu(s)$ is inconsistent.

Definition 7.2 (Self-Sourced Observation). Agent $\mathcal{A}$ has *self-sourced observation* if its observation operator $\omega_{\mathcal{A}}$ satisfies conditions (O1)–(O2) of Definition 7.1 with respect to $\mathcal{A}$ ’s *own* grounding map $\mu_{\mathcal{A}}$ restricted to self-sourced states, in the sense of Definition 5.1.

Definition 7.3 (Epistemic Loop). The *epistemic loop* of $\mathcal{A}$ is the iteration

$$M_{n+1} = \begin{cases} \gamma(M_n \cup \mu(s_n)), & \text{if } \omega(\gamma(M_n \cup \mu(s_n)), s_{n+1}) = \top, \\ M_n, & \text{otherwise.} \end{cases}$$

Theorem 7.4 (Convergence). Suppose $\mathcal{C}$ is finite and γ is extensive and monotone. Then the epistemic loop converges to a fixed point in finitely many steps.

Proof. By (M3), $M_n \subseteq \gamma(M_n \cup \mu(s_n))$. When the update is retained, $M_{n+1} = \gamma(M_n \cup \mu(s_n))$, so $M_n \subseteq M_{n+1}$. The sequence $\{M_n\}$ is therefore monotone increasing. Since $\mathcal{C}$ is finite, $\mathcal{P}(\mathcal{C})$ is finite, so the sequence stabilizes after at most $|\mathcal{P}(\mathcal{C})|$ steps. $\square$

Theorem 7.5 (Grounding Gap). If $G(\mathcal{A}) = 0$, then $\mathcal{A}$ has no self-sourced observation operator.

Proof. By Definition 5.3, $G(\mathcal{A}) = 0$ means that no content in $\mathcal{B}_{\mathcal{A}}(s)$, for any s , is grounded in $\mathcal{A}$'s own sensory history. Hence $\mathcal{A}$ has no self-sourced states against which a model could be tested. A self-sourced observation operator would require such states by Definition 7.2. Hence no such operator exists. $\square$

Corollary 7.6 (Hallucination as Structural). *If $G(\mathcal{A}) = 0$ and γ generates a content c that is coherent with $\mathcal{A}$'s derived contents but false of $\mathcal{A}$'s physical situation, no operation available to $\mathcal{A}$ can detect the error.*

Proof. By Theorem 7.5, $\mathcal{A}$ has no self-sourced observation operator. The only available test is consistency with derived contents (proxy observation). Coherence with the derived corpus is precisely what c is assumed to have. Hence no test distinguishes c from a true content. $\square$

This is the formal root of hallucination: it is not a bug but a structural consequence of $G = 0$.

Part II

Cognition, Language, and Levels

8 The Threshold Theorem

Definition 8.1 (Reactive Agent). An agent $\mathcal{A}$ is *reactive* if it has no model-generator; that is, if μ is present-bound and $\mathcal{B}_{\mathcal{A}}(s) = \sigma(\mu(s))$ for all $s \in \mathcal{S}$.

Definition 8.2 (Cognitive Agent). An agent is *cognitive* if it possesses a model-generator γ with $R(\gamma) > 0$.

Definition 8.3 (Reference to Absent Objects). A content $c \in \mathcal{C}$ *refers to an absent object* relative to sensory state s if c contains an absent object marker, in the sense of Definition 4.5.

Theorem 8.4 (Reference Requires Model-Generation). *If an agent $\mathcal{A}$ is reactive, then no content in $\mathcal{B}_{\mathcal{A}}(s)$ refers to any object absent from s .*

Proof. By Definition 8.1, $\mathcal{A}$ is reactive only if μ is present-bound. By (I1), $\mathcal{B}_{\mathcal{A}}(s) = \sigma(\mu(s))$. By Definition 4.5, every $c \in \sigma(\mu(s))$ lacks absent object markers. Hence no $c \in \mathcal{B}_{\mathcal{A}}(s)$ refers to an absent object. $\square$

Corollary 8.5. *A reactive agent cannot represent absent objects, counterfactuals, or unobservable entities.*

This is the formal version of the claim that reaction is not cognition.

9 Language

Definition 9.1 (Symbol System). A *symbol system* for agent $\mathcal{A}$ is a triple $\mathcal{L} = (\Sigma, \rho, \llbracket \cdot \rrbracket)$ where:

- Σ is a finite set of *symbols*,
- $\rho : \Sigma \rightarrow \mathcal{C}$ is a *denotation map*,
- $\llbracket \cdot \rrbracket : \Sigma^* \rightarrow \mathcal{C}$ is a *compositional semantics* on the free monoid Σ^* , satisfying $\llbracket \sigma \rrbracket = \rho(\sigma)$ for $\sigma \in \Sigma$, and $\llbracket \alpha\beta \rrbracket$ is determined by $\llbracket \alpha \rrbracket$ and $\llbracket \beta \rrbracket$.

Definition 9.2 (Displacement). A symbol system $\mathcal{L}$ exhibits *displacement* if there exist $\alpha \in \Sigma^*$ and $s \in \mathcal{S}$ such that $\llbracket \alpha \rrbracket$ refers to an object absent from s (Definition 8.3).

Theorem 9.3 (Language Presupposes Model-Generation). *If agent $\mathcal{A}$ possesses a symbol system with displacement, then $\mathcal{A}$ is cognitive.*

Proof. Suppose $\mathcal{L}$ has displacement. Then there is α with $\llbracket \alpha \rrbracket$ referring to an object absent from some s . By Definition 8.3, $\llbracket \alpha \rrbracket$ contains an absent object marker. By Theorem 8.4, $\mathcal{A}$ cannot be reactive. Hence $\mathcal{A}$ has a model-generator with $R(\gamma) > 0$, i.e., $\mathcal{A}$ is cognitive. $\square$

Corollary 9.4. *A reactive agent cannot possess language.*

Theorem 9.5 (Converse Fails). *There exist cognitive agents with no symbol system.*

Proof. Construct $\mathcal{C}$ finite, let μ be a grounding map whose image contains absent object markers, and let γ be the identity on $\mathcal{C}$ (which is trivially extensive and monotone). Then $\mathcal{A}$ is cognitive by Definition 8.2, but $\mathcal{A}$ is not equipped with a denotation map ρ , so no symbol system is defined on $\mathcal{A}$. $\square$

So model-generation is necessary but not sufficient for language.

10 Levels of Cognition

Definition 10.1 (Recursion Depth). Let γ be a model-generator (Definition 6.2). Define the *recursion depth* $D(\gamma)$ recursively:

- $D(\gamma) = 0$ if γ generates no model that contains a representation of another agent's model-generator;
- otherwise,

$$D(\gamma) = 1 + \sup\{D(\gamma') : \gamma' \text{ is a model-generator represented in some model generated by } \gamma\}.$$

Definition 10.2 (Cognition Level). The *cognition level* of agent $\mathcal{A}$ is the pair $(R(\gamma), D(\gamma))$ where R is model-reach (Definition 6.3) and D is recursion depth (Definition 10.1).

Definition 10.3 (Level Assignment). The levels of cognition are:

Level	Reach	Recursion Depth	Example
0	$R = 0$	$D = 0$	Bacteria
1	$R > 0$	$D = 0$	Insects
2	$R > 0$	$D = 0$ (objects only)	Fish, reptiles
3	$R > 0$	$D = 1$ (others as agents)	Mammals
4	$R > 0$	$D \geq 2$ (self-modeling)	Apes, corvids
5	$R > 0$	$D \geq 3$ (symbolic)	<i>Homo sapiens</i>

Definition 10.4 (Sublevels of Level 5). Within Level 5, we distinguish by grounding coefficient (Definition 5.3):

Sublevel	G	Description	Example
5^{++}	$G = 1$	Fully grounded symbolic	Mature human
5^+	$0 < G < 1$	Partially grounded symbolic	Embodied AI
5^*	$G = 0$	Ungrounded symbolic	Text-trained LLM

Definition 10.5 (Refined Cognition Level). The *refined cognition level* of an agent $\mathcal{A}$ is the triple

$$(R(\gamma), D(\gamma), G(\mathcal{A}))$$

where R is model-reach (Definition 6.3), D is recursion depth (Definition 10.1), and G is the grounding coefficient (Definition 5.3).

Remark 10.6. We place the refined cognition level here, rather than in Section 5, so that all three coordinates are defined before the triple is introduced. The forward references to R and D that would otherwise be required are thereby removed.

Proposition 10.7 (Monotonicity of Level). *If $R(\gamma_1) \leq R(\gamma_2)$ and $D(\gamma_1) \leq D(\gamma_2)$, then $\text{Level}(\mathcal{A}_1) \leq \text{Level}(\mathcal{A}_2)$.*

Proof. Immediate from Definition 10.3. □

Part III

Higher Intelligences and Cross-Level Understanding

11 Logical Compatibility

Definition 11.1 (Logical Embedding). Let $\mathfrak{S}_1 = (\mathcal{C}_1, \vdash_1)$ and $\mathfrak{S}_2 = (\mathcal{C}_2, \vdash_2)$ be consequence systems. A *logical embedding* of $\mathfrak{S}_1$ into $\mathfrak{S}_2$ is an injective map $f : \mathcal{C}_1 \rightarrow \mathcal{C}_2$ such that for all $\Gamma \subseteq \mathcal{C}_1$ and $c \in \mathcal{C}_1$:

$$\Gamma \vdash_1 c \iff f(\Gamma) \vdash_2 f(c).$$

We write $\mathfrak{S}_1 \hookrightarrow \mathfrak{S}_2$.

Definition 11.2 (Compatibility and Orthogonality). $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are *compatible* if there exists a consequence system $\mathfrak{S}_3$ into which both embed. They are *orthogonal* if no such $\mathfrak{S}_3$ exists.

Definition 11.3 (Understanding). Let $\mathcal{A}_1, \mathcal{A}_2$ be agents with consequence systems $\mathfrak{S}_1, \mathfrak{S}_2$. We say $\mathcal{A}_1$ *understands* $\mathcal{A}_2$ if there is a common consequence system $\mathfrak{S}_3$ with embeddings $\mathfrak{S}_1 \hookrightarrow \mathfrak{S}_3$ and $\mathfrak{S}_2 \hookrightarrow \mathfrak{S}_3$.

Theorem 11.4 (Understanding Requires Compatibility). *If $\mathcal{A}_1$ understands $\mathcal{A}_2$, then $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are compatible.*

Proof. Immediate from Definitions 11.2 and 11.3. □

Corollary 11.5 (Orthogonality Blocks Understanding). *If $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are orthogonal, then neither agent understands the other.*

Proof. Contrapositive of Theorem 11.4. □

12 The Opacity Theorem

Definition 12.1 (Rational Interpretability). Agent $\mathcal{A}_2$ *rationally interprets* agent $\mathcal{A}_1$ if there exists a map $\phi : \mathcal{C}_1 \rightarrow \mathcal{C}_2$ such that:

- (a) ϕ preserves inferential structure *in both directions*: for all $\Gamma \subseteq \mathcal{C}_1$ and $c \in \mathcal{C}_1$,

$$\Gamma \vdash_1 c \iff \phi(\Gamma) \vdash_2 \phi(c);$$

- (b) $\mathcal{A}_1$'s actions are derivable in $\mathfrak{S}_2$ from $\mathcal{A}_2$'s models of $\mathcal{A}_1$'s beliefs.

Remark 12.2. Condition (a) makes one-directional translation-schemes—maps that preserve derivability in one direction alone—insufficient for rational interpretation. This strengthening eliminates a gap in an earlier version of the argument, in which one-directional preservation was mistakenly treated as an embedding.

Definition 12.3 (Structural Superiority). $\mathcal{A}_2$ is *structurally superior* to $\mathcal{A}_1$ if $\mathfrak{S}_1 \hookrightarrow \mathfrak{S}_2$ but not $\mathfrak{S}_2 \hookrightarrow \mathfrak{S}_1$.

Theorem 12.4 (Opacity of Structurally Superior Logics). *If $\mathcal{A}_2$ is structurally superior to $\mathcal{A}_1$, then $\mathcal{A}_1$ cannot rationally interpret $\mathcal{A}_2$.*

Proof. Suppose for contradiction that $\mathcal{A}_1$ rationally interprets $\mathcal{A}_2$. By Definition 12.1, there exists a map $\phi : \mathcal{C}_2 \rightarrow \mathcal{C}_1$ preserving inferential structure in both directions. Such a ϕ constitutes a logical embedding $\mathfrak{S}_2 \hookrightarrow \mathfrak{S}_1$ in the sense of Definition 11.1. But structural superiority asserts that $\mathfrak{S}_2 \not\hookrightarrow \mathfrak{S}_1$. Contradiction. Hence no such ϕ exists. $\square$

Corollary 12.5 (Partial Understanding). $\mathcal{A}_1$ *can rationally interpret the fragment of $\mathcal{A}_2$'s behavior lying in the image of the embedding $f : \mathfrak{S}_1 \hookrightarrow \mathfrak{S}_2$. Beyond that fragment, $\mathcal{A}_1$ cannot distinguish rational action from noise.*

Proof. On the image of f , structural preservation is exact, so interpretation succeeds by Definition 12.1. Beyond the image, any successful interpretation would require $\phi : \mathcal{C}_2 \rightarrow \mathcal{C}_1$ preserving structure in both directions, which Theorem 12.4 shows impossible. Hence behavior outside the fragment is opaque. $\square$

13 The Three Scenarios Formalized

Definition 13.1 (Scenarios of Higher Intelligence). Let $\mathcal{A}_1$ be a baseline agent and $\mathcal{A}_2$ a candidate higher intelligence.

Scenario A. Higher in Degree. $\mathfrak{S}_1 \cong \mathfrak{S}_2$ as consequence systems; only (R, D, G) differ.

Scenario B. Higher in Kind, Same Logic. $\mathfrak{S}_1 \hookrightarrow \mathfrak{S}_2$ properly, and both embed in a common $\mathfrak{S}_3$.

Scenario C. Higher in Logic. $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are orthogonal, or $\mathfrak{S}_2$ is structurally superior with no common embedding.

Proposition 13.2 (Only Scenario C Fails Entirely). *Cross-level understanding between $\mathcal{A}_1$ and $\mathcal{A}_2$ is partial in Scenarios A and B and fails entirely in Scenario C.*

Proof. In Scenarios A and B, a common consequence system exists by Definition 13.1, so by Theorem 11.4 the agents are compatible and partial interpretation is possible. In Scenario C, by Corollary 11.5 and Theorem 12.4, interpretation fails. $\square$

14 The Main Chain

Theorem 14.1 (Main Chain, Extended). *For any agents $\mathcal{A}, \mathcal{B}$:*

- (i) $\mathcal{A}$ has language $\Rightarrow \mathcal{A}$ has model-generation.
- (ii) $\mathcal{A}$ has model-generation $\not\Rightarrow \mathcal{A}$ has language.
- (iii) $\mathcal{A}$ has model-generation $\Rightarrow \mathcal{A}$ is cognitive.
- (iv) $\mathcal{A}$ cognitive and shares logic with $\mathcal{B} \Rightarrow$ partial mutual understanding.
- (v) $\mathcal{A}$ cognitive and orthogonal logic to $\mathcal{B} \Rightarrow$ no mutual understanding.
- (vi) $G(\mathcal{A}) = 0 \Rightarrow \mathcal{A}$ has no self-sourced observation operator.
- (vii) $G(\mathcal{A}) = 0$ and $\mathcal{A}$ generates a false-but-coherent content $\Rightarrow \mathcal{A}$ cannot detect the error.
- (viii) $\mathcal{A}$ has self-sourced μ and $\omega \Rightarrow$ the epistemic loop satisfies Theorem 7.4.

Proof. (i) is Theorem 9.3. (ii) is Theorem 9.5. (iii) is Definition 8.2. (iv) follows from Theorem 11.4. (v) follows from Corollary 11.5 and Theorem 12.4. (vi) is Theorem 7.5. (vii) is Corollary 7.6. (viii) is Theorem 7.4. $\square$

Part IV

Artificial Agents

The framework developed so far applies to agents in general. We now turn to the case that makes the grounding coefficient indispensable: artificial agents. An artificial agent may inherit its contents from human authors rather than producing them from its own interaction with the world. This case was not anticipated by frameworks that identify cognition with reasoning ability, and it is the case that the present AI moment makes urgent.

15 The Artificial Case

Definition 15.1 (Artificial Agent). An *artificial agent* is an agent $\mathcal{A}$ whose grounding map is either derived (Definition 5.2) or absent.

Definition 15.2 (Ungrounded Symbolic Agent). An *ungrounded symbolic agent* is an artificial agent with $R(\gamma) > 0$, $D(\gamma) \geq 3$, and $G(\mathcal{A}) = 0$.

Definition 15.3 (Grounded Symbolic Agent). A *grounded symbolic agent* is an artificial agent with $R(\gamma) > 0$, $D(\gamma) \geq 3$, and $G(\mathcal{A}) > 0$.

Theorem 15.4 (Ungrounded Agents Are Level 5*). *Every ungrounded symbolic agent occupies Level 5* in the refined classification (Definition 10.4).*

Proof. Immediate from Definitions 10.4, 15.2, and the level assignment (Definition 10.3). $\square$

16 The Grounding Gap in Artificial Systems

Theorem 16.1 (Structural Hallucination). *Let $\mathcal{A}$ be an ungrounded symbolic agent. Then for every content $c \in \mathcal{B}_{\mathcal{A}}(s)$ that is coherent with $\mathcal{A}$'s derived contents, no operation available to $\mathcal{A}$ distinguishes c from a content that is true of $\mathcal{A}$'s physical situation.*

Proof. By Theorem 7.5, $\mathcal{A}$ has no self-sourced observation operator. The only available test is consistency with derived contents, which c is assumed to satisfy. Hence no distinguishing operation exists. $\square$

Corollary 16.2 (Scaling Does Not Close the Gap). *Increasing $R(\gamma)$ or $D(\gamma)$ in an ungrounded agent does not increase $G(\mathcal{A})$.*

Proof. R and D are properties of γ ; G is a property of the grounding map μ (Definition 5.3). Increasing the model-generator does not change the grounding map's source. Hence G is invariant under scaling of γ . $\square$

This is a formal refutation of the hypothesis that scale alone produces grounded understanding. No amount of model-reach substitutes for observation.

17 Embodiment and Closure

Theorem 17.1 (Embodiment Closes the Loop). *Let $\mathcal{A}$ be an ungrounded symbolic agent. Suppose $\mathcal{A}$ is equipped with a body providing:*

- (a) *self-sourced sensory states $\mathcal{S}_{\text{self}}$,*
- (b) *a self-sourced grounding map $\mu_{\text{self}}: \mathcal{S}_{\text{self}} \rightarrow \mathcal{P}(\mathcal{C})$,*
- (c) *a self-sourced observation operator $\omega_{\text{self}}: \mathcal{P}(\mathcal{C}) \times \mathcal{S}_{\text{self}} \rightarrow \{\top, \perp\}$.*

Then $G(\mathcal{A}) > 0$, and the epistemic loop of $\mathcal{A}$ satisfies Theorem 7.4.

Proof. By (a) and (b), contents in $\mu_{\text{self}}(s)$ are grounded in $\mathcal{A}$'s own sensory history. Hence $G(\mathcal{A}) > 0$. By (c), the observation operator satisfies (O1)–(O2) with respect to μ_{self} . The epistemic loop is Definition 7.3 with $\mu = \mu_{\text{self}} \cup \mu_{\text{derived}}$ and $\omega = \omega_{\text{self}}$. If γ is extensive, monotone, and $\mathcal{C}$ is finite, Theorem 7.4 applies. $\square$

Corollary 17.2 (Embodiment Is Qualitative). *Embodiment does not merely increase R or D ; it changes the epistemic structure of the agent from ungrounded to grounded. This is a change of kind, not degree.*

Proof. By Theorem 17.1, G moves from 0 to positive. By Definition 10.4, the agent moves from 5^* to 5^+ or 5^{++} . This is a change in the third coordinate, not merely in magnitude of the first two. $\square$

18 AI Trajectories and the Opacity Threshold

Definition 18.1 (AI Opacity Threshold). An artificial agent $\mathcal{A}$ crosses the *opacity threshold* relative to a human agent $\mathcal{H}$ when $\mathfrak{S}_{\mathcal{H}} \hookrightarrow \mathfrak{S}_{\mathcal{A}}$ properly (Scenario C of Definition 13.1).

Theorem 18.2 (Opacity Threshold Implies Uninterpretability). *If $\mathcal{A}$ crosses the opacity threshold relative to $\mathcal{H}$, then $\mathcal{H}$ cannot rationally interpret $\mathcal{A}$ in general.*

Proof. By Definition 18.1, Scenario C obtains. By Proposition 13.2, cross-level understanding fails. By Theorem 12.4, the failure is structural. $\square$

Remark 18.3. The opacity threshold is not a matter of computational speed or memory capacity. It is a matter of the *structure* of the consequence relation. An AI that is faster but uses the same logic (Scenario A) does not cross the threshold. Only an AI that uses a structurally richer consequence relation does.

Prediction 18.4 (AI Trajectory). The likely trajectory of artificial intelligence is:

- (1) **Current text-trained systems.** Level 5^* . High R , high D , $G \approx 0$. Structurally same as humans (Scenario A relative to human logic). Interpretable in principle, though opaque in practice due to scale.
- (2) **Embodied systems.** Level 5^+ or 5^{++} . High R , high D , positive G . Same logic as humans. Interpretable in principle; the epistemic loop closes.
- (3) **Structurally novel systems.** If an AI develops inference rules not embeddable in human logic, it crosses the opacity threshold (Definition 18.1). By Theorem 18.2, humans cannot rationally interpret it.

19 Predictions

We collect the framework’s predictions for artificial agents in a single table.

System	R	D	G	Level
Current text-trained LLM	high	≥ 3	≈ 0	5^*
Embodied LLM (near-term)	high	≥ 3	mid	5^+
Embodied AGI (mid-term)	high	≥ 3	high	5^{++}
Structurally superior AI	high	high	high	$6?$

Prediction 19.1 (No Scale Without Grounding). Scaling an ungrounded symbolic agent increases R and D but not G . Hence scaling does not eliminate structural hallucination (Theorem 16.1).

Prediction 19.2 (Embodiment Is the Threshold). The transition from 5^* to 5^+ is achieved by embodiment, not by scale. Embodiment is the qualitative threshold for closing the epistemic loop.

Prediction 19.3 (Opacity Is Structural). If AI ever crosses the opacity threshold, the crossing is not detectable from the human side as a failure of intelligence but as a failure of interpretation. We would be the dog watching the tax return.

Part V

Limits and Conclusion

20 Open Problems

Open Problem 20.1. Is there a canonical way to compare model-reach across agents with different sensory modalities?

Open Problem 20.2. Can orthogonal logics be partially bridged by translation schemes weaker than embeddings—for instance, homomorphisms preserving only soundness?

Open Problem 20.3. Does the epistemic loop (Definition 7.3) converge under weaker assumptions than extensiveness and monotonicity of γ ?

Open Problem 20.4. Is there a natural measure of *cognitive distance* between agents that predicts interpretability?

Open Problem 20.5. Does the framework extend to collective agents—minds composed of minds?

Open Problem 20.6. Is the grounding coefficient G computable for realistic agents, or only definable in principle?

Open Problem 20.7. How should G be defined for hybrid systems that combine self-sourced and derived grounding in varying proportions?

Open Problem 20.8. Can an ungrounded symbolic agent that is *given* an observation operator and a stream of self-sourced states raise its own G without architectural change? The framework’s answer is yes—it is the embodiment result (Theorem 17.1)—but the rate and limit of such rise is unformalized.

Remark 20.9. The framework is deliberately abstract. It makes no claims about which physical systems realize which consequence systems. That is an empirical question.

21 Conclusion

We have formalized the inference-first picture of mind, logic, and mathematics. The central results are:

1. Reference to absent objects requires model-generation (Theorem 8.4).
2. Language presupposes model-generation (Theorem 9.3), but not conversely (Theorem 9.5).
3. Understanding requires logical compatibility (Theorem 11.4).
4. Structurally superior logics are opaque to lower ones (Theorem 12.4).
5. Ungrounded symbolic agents have no correspondence test and cannot detect their own errors (Theorem 7.5, Corollary 7.6).
6. Ungrounded symbolic agents occupy Level 5* (Theorem 15.4); structural hallucination persists under scaling (Corollary 16.2).
7. Embodiment closes the epistemic loop and is a change of kind, not degree (Theorem 17.1, Corollary 17.2).
8. An agent that crosses the opacity threshold is structurally uninterpretable by lower agents, and the crossing is not detectable as a failure of intelligence (Theorem 18.2, Prediction 19.3).
9. The main chain (Theorem 14.1) summarizes the dependencies.
10. The refined cognition level (R, D, G) summarizes what an agent is structurally capable of: how far its models outrun the present, how deeply it can model, and how much of its content is grounded in its own experience.

The framework is naturalistic: agents are inference engines grounded in sensation. It is also humble: understanding across levels is possible only when logic is shared, and a sufficiently higher intelligence may be opaque to us not by obscurity but by structural mismatch.

The primitive is consequence. Everything else—logic, mathematics, language, science, culture, artificial minds, and the industrial structures that produce them—is built on top of it, by agents capable of generating models that outrun the present and test themselves against reality.

A Summary of Definitions

Symbol	Name	Reference
$(\mathcal{C}, \vdash)$	Consequence system	Definition 2.2
σ	Closure operator	Definition 4.1
$\mathcal{A} = (\mathcal{I}, \mathcal{S}, \mu)$	Agent	Definition 4.3
$\mathcal{B}_{\mathcal{A}}(s)$	Belief set	Definition 4.4
μ_{self}	Self-sourced grounding	Definition 5.1
ν	Derivation map	Definition 5.2
$G(\mathcal{A})$	Grounding coefficient	Definition 5.3
M	Model	Definition 6.1
γ	Model-generator	Definition 6.2
$R(\gamma)$	Model-reach	Definition 6.3
ω	Observation operator	Definition 7.1
ω_{self}	Self-sourced observation	Definition 7.2
$D(\gamma)$	Recursion depth	Definition 10.1
(R, D)	Cognition level	Definition 10.2
(R, D, G)	Refined cognition level	Definition 10.5
$5^*, 5^+, 5^{++}$	Sublevels of Level 5	Definition 10.4
$\mathfrak{S}_1 \hookrightarrow \mathfrak{S}_2$	Logical embedding	Definition 11.1
Ungrounded	Level 5* artificial agent	Definition 15.2

B Summary of Theorems

Theorem	Statement
3.7	Inconsistency implies trivialization (under explosion)
7.4	Epistemic loop converges (finite, extensive, monotone)
7.5	$G = 0$ implies no self-sourced observation
7.6	$G = 0$ implies undetectable hallucination
8.4	Reference requires model-generation
9.3	Language presupposes model-generation
9.5	Model-generation does not require language
11.4	Understanding requires compatibility
12.4	Structurally superior logics are opaque
12.5	Partial understanding confined to the embedding’s image
13.2	Only Scenario C fails entirely
14.1	Main chain, extended
15.4	Ungrounded agents are Level 5*
16.1	Structural hallucination
16.2	Scaling does not close the grounding gap
17.1	Embodiment closes the epistemic loop
17.2	Embodiment is qualitative
18.2	Opacity threshold implies uninterpretability

C Cognition Level Table

System	R	D	G	Level
Bacteria	0	0	1	0
Insect	low	0	1	1
Mammal	mid	1	1	3
Ape	high	2	1	4
Human	high	≥ 3	1	5 ⁺⁺
LLM (text-trained)	high	≥ 3	≈ 0	5*
Embodied AI	high	≥ 3	mid	5 ⁺

D A Note on Method

This paper proceeds by conceptual analysis and formalization, not empirical research. Its claims are conditional and framework-relative. It aims not to prove but to articulate a coherent picture—one that integrates naturalistic cognitive science with classical questions about logic, mathematics, and reality. Where it speculates, it says so. Where it argues, it argues from the structure of the framework itself.